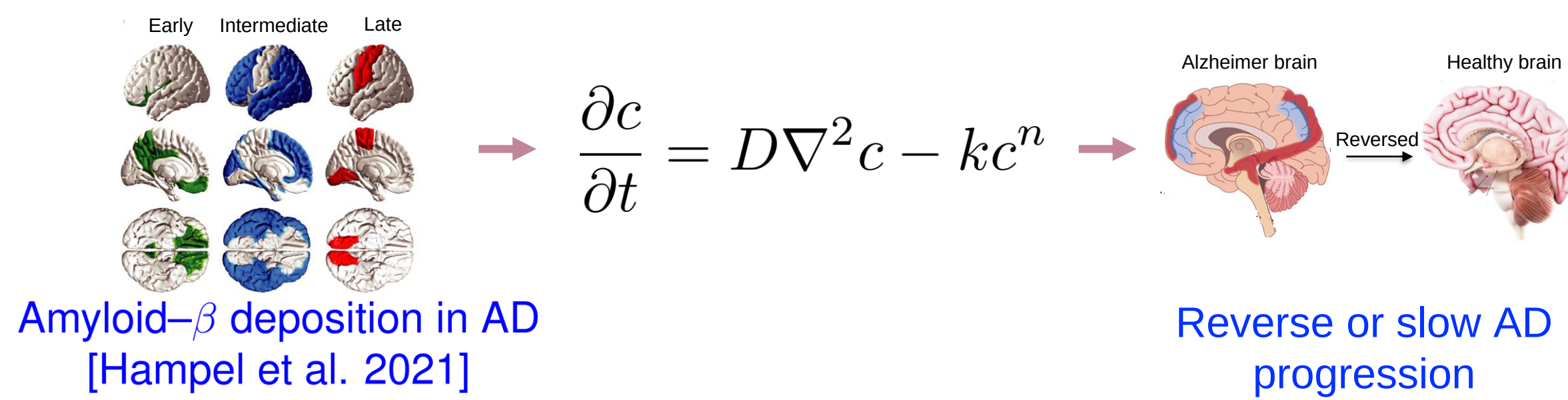
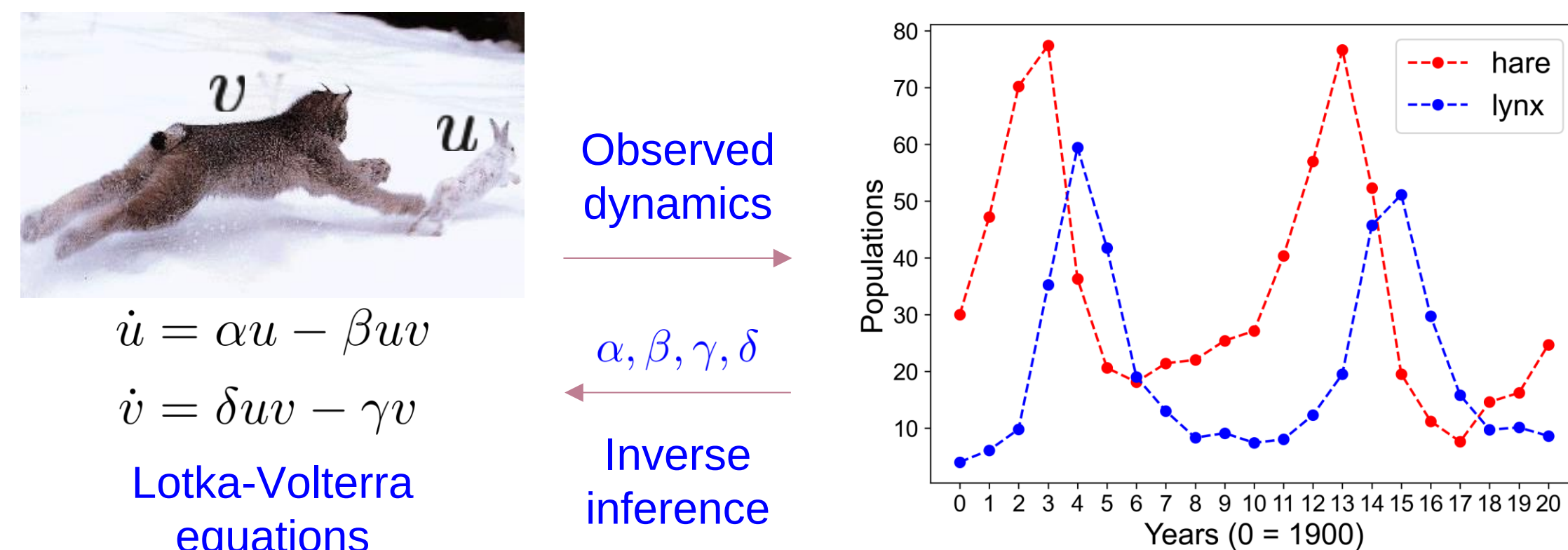


Discovering physical laws



Physical laws are critical to predicting/controlling complex systems, but traditional first-principles derivations may be infeasible

Discovering physics from data: Example



General equation of a dynamical system

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t))$$

State variable ← → Governing dynamics

Objective

- The **objective** of physics discovery is to infer $f(\mathbf{x}(t))$ from **data** $\mathbf{x}(t)$

Sparse Regression [Brunton et al.]

- Reformulate the physics using a “library” of physics

$$\frac{d}{dt}\mathbf{x}(t) = f(\mathbf{x}(t)) \longrightarrow \dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi$$

Library of functions → Unknown coefficients

- Example of physics library for Lotka Volterra –

$$\begin{aligned} \dot{u} &= \alpha u - \beta uv \\ \dot{v} &= \delta uv - \gamma v \end{aligned} \longrightarrow \begin{bmatrix} \dot{u} \\ \dot{v} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & u & v & u^2 & v^2 & uv \end{bmatrix}}_{\Theta(u,v)} \begin{bmatrix} \xi_{11} & \xi_{12} \\ \xi_{21} & \xi_{22} \\ \xi_{31} & \xi_{32} \\ \xi_{41} & \xi_{42} \\ \xi_{51} & \xi_{52} \\ \xi_{61} & \xi_{62} \end{bmatrix}$$

- Physics discovery using sparse regression

$$\Xi^* = \arg \min_{\Xi} \left\| \dot{\mathbf{X}} - \Theta(\mathbf{X})\Xi \right\|_F^2 + \Omega(\Xi)$$

Noisy measurements → Regularization

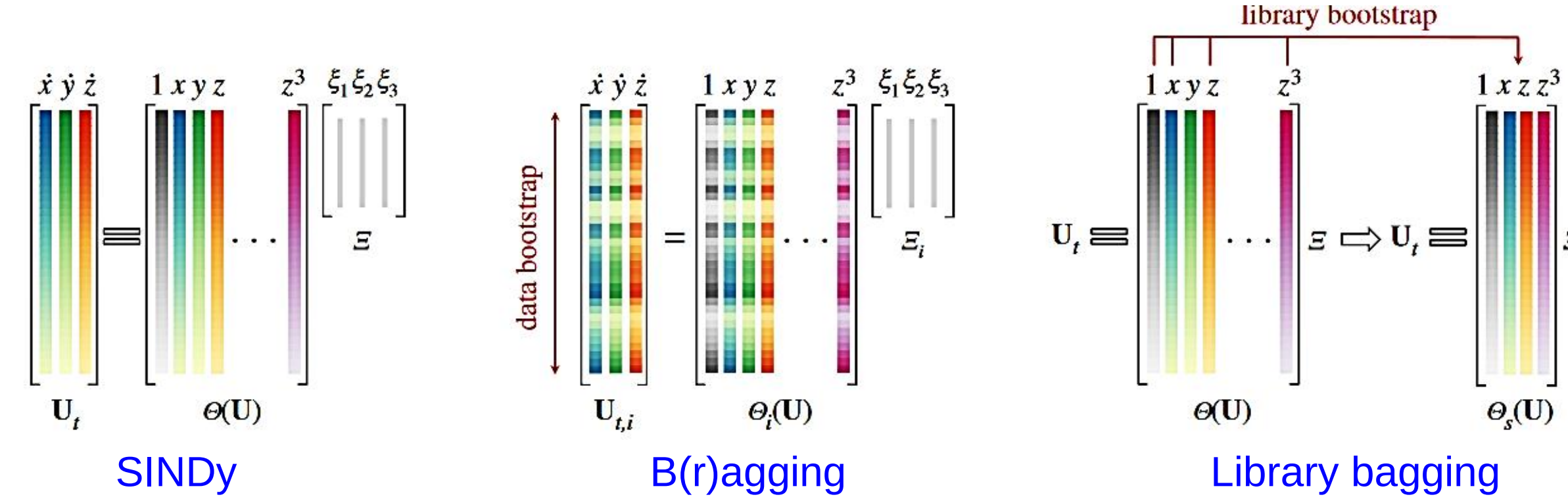
Ground truth (from literature)	$\dot{u} = 0.55u - 0.45uv$ $\dot{v} = -0.84v + 0.543uv$
Discovered dynamics	$\dot{u} = -0.31 + 0.49u - 0.36uv$ $\dot{v} = 0.14 + 0.21u - 1.11v + 0.31uv + 0.16v^2$

Limitations

- Retains spurious higher-order terms and omit crucial physics
- Highly sensitive to noise leading to poor generalization

Ensemble SINDy [Fasel et al., 2022]:

- Employs data/library bootstrapping to recover multiple SINDy models



Ground truth (from literature)	$\dot{u} = 0.55u - 0.45uv$ $\dot{v} = -0.84v + 0.543uv$
Discovered dynamics	$\dot{u} = 0.12 + 0.48u - 0.51uv$ $\dot{v} = 0.14 + 0.33u - 0.81v + 0.50uv$

Limitations

- Also incorrectly learns extra (higher order) terms
- Estimated coefficients are inaccurate

Can we eliminate extraneous terms from the learned model?

CaNDiCE for Physics Discovery

- We adopt counterfactual explanation to identify and eliminate spurious physics terms from the governing equation

Data model and physics discovery

- Given the reformulated physics, we account for the measurement noise through observation

$$\mathbf{Y} = \int \dot{\mathbf{X}} dt + \epsilon = \int \Theta(\mathbf{X})\Xi dt + \epsilon$$

Measurement noise $\epsilon \sim N(0, \sigma_{\text{noise}}^2)$

- Sparse identification is used to estimate an initial governing physics

$$\hat{\mathcal{Q}}(\hat{\Xi}) = \Theta(\mathbf{X})\hat{\Xi}$$

Contains nonzero coefficient for spurious terms

Distributional counterfactual explanation (Sparse CounterGAN)

Objective: Isolate the true causal physics from the over-parameterised governing equation

- We define a distributional classifier (C) that labels coefficients (μ_{Ξ}) as in-distribution or out of distribution using Kolmogorov-Smirnov test

- The labels drive a Generator-Discriminator (SCGAN) pair trained to identify spurious terms by optimizing the three (3) objectives below. G adds residual $\mathbf{r} = G(\mu_{\Xi})$ to flip the label, while D enforces realism.

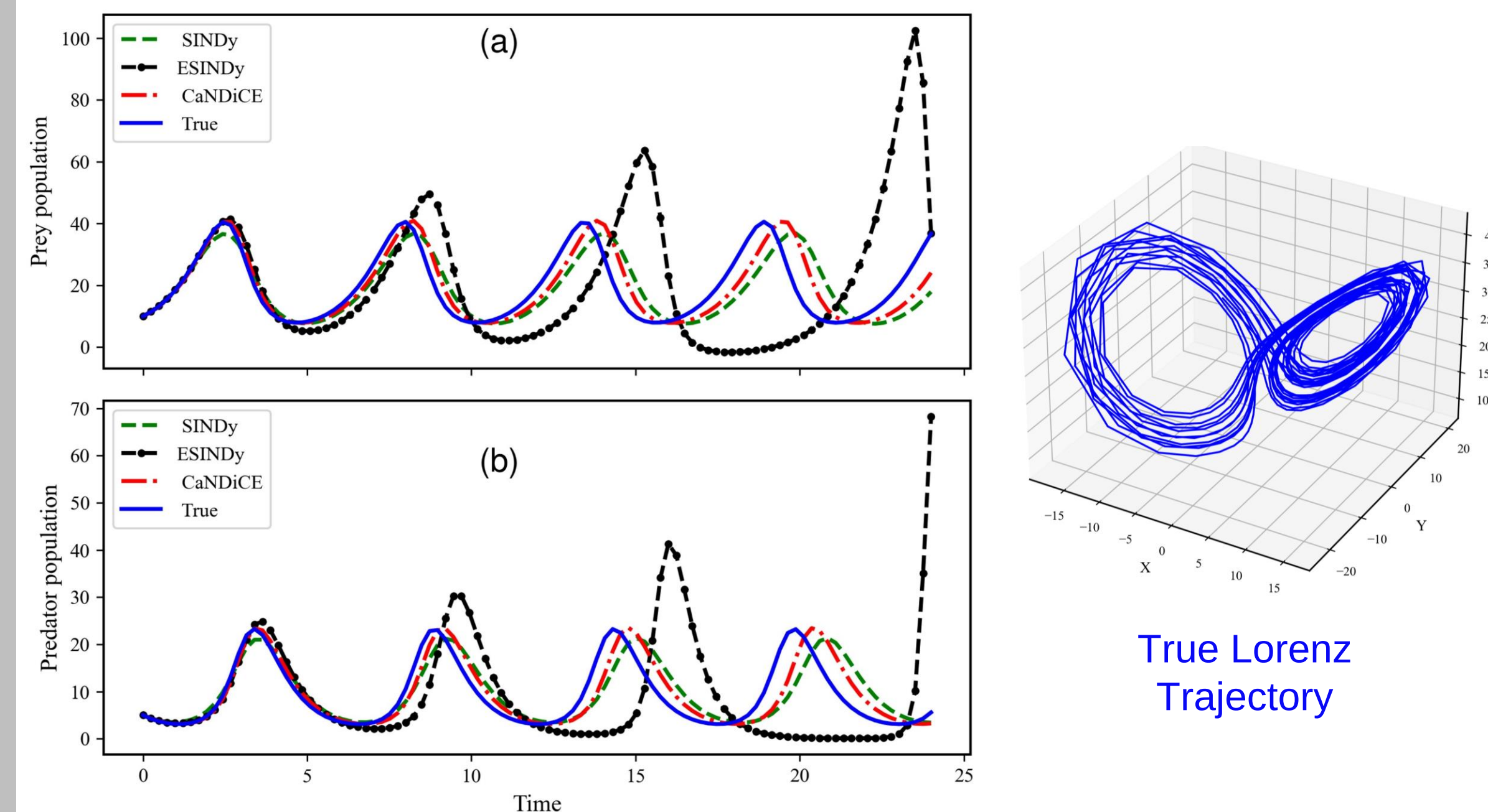
- $L_{\text{GAN}}(G, D) = \mathbb{E}_{\mathbf{x}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{x}}[\log (1 - D(\mathbf{x} + G(\mathbf{x})))]$ (Adversarial loss)
- $L_c(G, C) = \mathbb{E}_{(\mathbf{x}, y)}[-y \log C(\mathbf{x} + G(\mathbf{x})) - (1 - y) \log(1 - C(\mathbf{x} + G(\mathbf{x})))]$ (Entropy loss)
- $L_{\text{Reg}}(G) = \alpha \sum_i \|G(\mathbf{x}_i)\|_1 - \beta \frac{1}{|\mathbf{X}|} \sum_{\mathbf{x} \in \mathbf{X}} \min_{\mathbf{s} \in S} \left(\sum_{i=1}^{N_f} (c_{\mathbf{x}}^i - s^i)^2 \right)^{1/2}$ (Sparsity loss) (Diversity loss)

SCGAN output: Pruned causal library $\tilde{\Theta}(\mathbf{X})$ composed of the minimal set of essential terms. This is used to recover the physics:

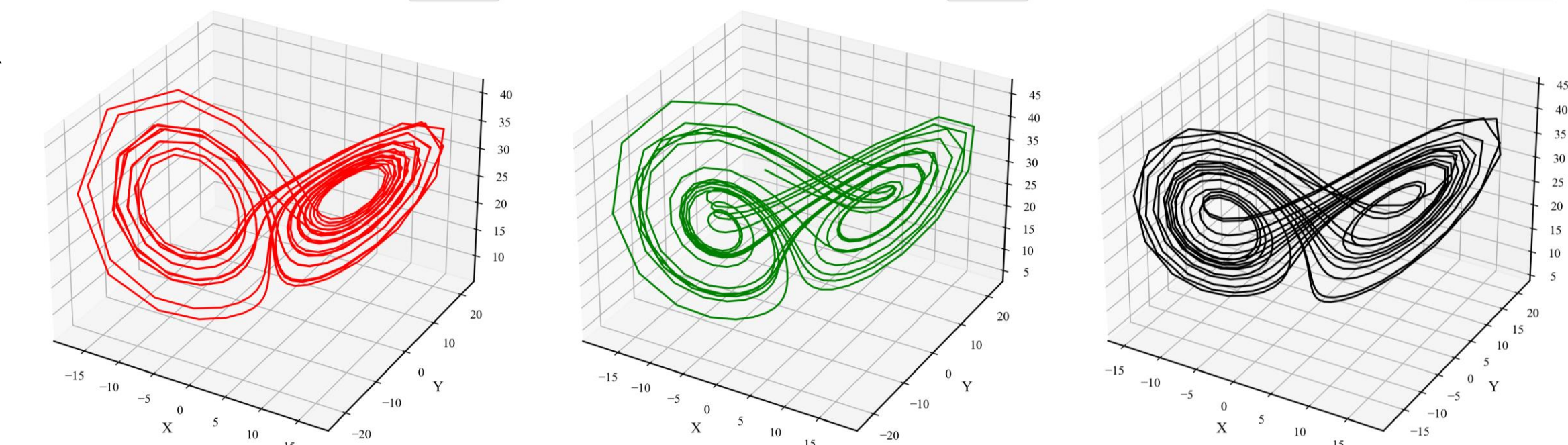
$$\hat{\mathcal{Q}}(\hat{\Xi}) = \tilde{\Theta}(\mathbf{X})\hat{\Xi}$$

Case Study

Model / Method	Lotka–Volterra	Lorenz
True Physics	$\dot{u} = 1.0u - 0.1uv$ $\dot{v} = -1.5v + 0.075uv$	$\dot{x} = -10x + 10y$ $\dot{y} = 28x - y - xz$ $\dot{z} = -2.67z + xy$
CaNDiCE	$\dot{u} = 0.98u - 0.097uv$ $\dot{v} = -1.45v + 0.072uv$	$\dot{x} = -10.13x + 9.96y$ $\dot{y} = 26.4x - 1.03y - 0.98xz$ $\dot{z} = -2.85z + 0.93xy$
SINDy	$\dot{u} = 2.9 + 0.84u - 0.43v - 0.074uv$ $\dot{v} = -0.77 + 0.08u - 1.26v + 0.06uv$	$\dot{y} = 1.24 + 9.95x + 3.90y - 0.50xz$ $\dot{z} = -12.60 - 0.05x - 1.88z + 0.91xy$ $\dot{x} = -9.31x + 9.32y$
ESINDy	$\dot{u} = 2.8 + 0.83u - 0.44v - 0.073uv$ $\dot{v} = -1.334v + 0.066uv$	$\dot{y} = 3.19 + 8.04x + 3.24y - 0.54xz$ $\dot{z} = -19.36 - 0.75x + 0.65y - 1.56z + 0.89xy$



Lotka-Volterra Trajectories Comparison



CaNDiCE Trajectory

SINDy Trajectory

ESINDy Trajectory

System		CaNDiCE	SINDy	ESINDy	
		RMSE	RMSE	% error reduction	% error reduction
Lotka–Volterra	Model Coeffs	0.02	0.87	98%	0.85
	Trajectory	4.67	6.53	28.57%	18.54
Lorenz	Model Coeffs	0.30	4.45	93.3%	3.72
	Trajectory	10.81	11.88	9.03%	12.66

Summary

- Introduced CaNDiCE, an explainable physics discovery framework.
- Achieves state-of-the-art accuracy on benchmarked systems, outperforming SINDy (93%) and ESINDy (92%).
- Provides causal interpretability & enhances trust in data-driven physics discovery.

Future work

- Refined classifier for improved distribution identification
- Extend CaNDiCE framework to address missing physics terms cases

References

- Brunton et al., Proceedings of the National Academy of Sciences, 113. 3932-3937, 2016
- Fasel et al, Proceedings of the Royal Society A 478.2260 (2022): 20210904